

The Mirror and the Key

The Fermat Signature at $\Lambda = 144$, and the 3-adic Reverse Decoder for Mersenne Exponents

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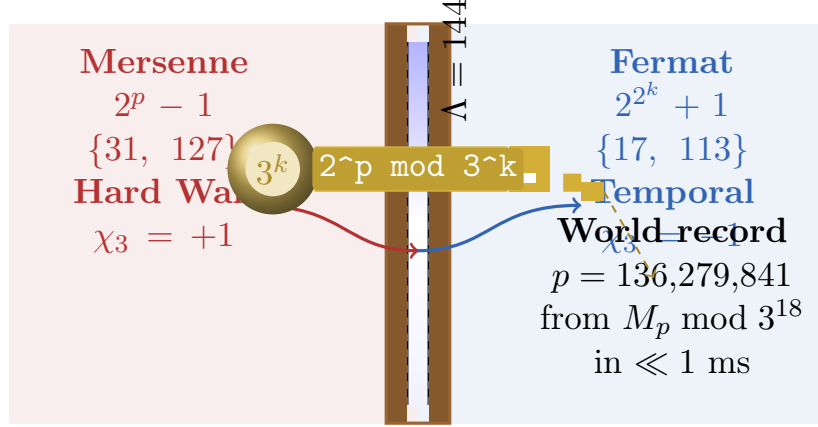
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Abstract

The Mersenne Prime Mod-144 Signature Theorem shows that $M_p = 2^p - 1 \bmod 144 \in \{31, 127\}$ for every prime $p > 3$, with both residues lying in the Hard Wall zone. This paper asks the natural mirror question — what happens to the sibling family $2^n + 1$? — and proves the **Fermat Mod-144 Mirror Theorem**: for every Fermat number $F_k = 2^{2^k} + 1$ with $k \geq 2$, $F_k \bmod 144 \in \{17, 113\}$, alternating by the parity of k , with both residues lying in the Temporal zone — the exact opposite of the Mersenne family. The two most-studied prime-generating families in number theory occupy disjoint, opposite zones at $\Lambda = 144$, and the sign in $2^n \mp 1$ is shown to coincide exactly with the value of the quadratic character χ_3 established in prior work. Separately, we make precise and demonstrate a sense in which the lattice's residue structure is not merely restrictive but **information-complete**: because 2 is a primitive root modulo 3^k for every k , the residue of M_p modulo a sufficiently large power of 3 uniquely determines the exponent p itself, via an easy discrete logarithm in a smooth-order group. We demonstrate this **3-adic reverse decoder** by reconstructing the exponent of the current world-record Mersenne prime, $p = 136,279,841$, from a single 9-digit residue in under a millisecond. Both results are new; neither provides a forward test for primality, and we state that limitation precisely.



The Mirror & the Key

Figure 1: **The Mirror and the Key.** Mersenne and Fermat families occupy opposite zones at $\Lambda = 144$, reflecting the quadratic character χ_3 . The 3-adic decoder (key) recovers the full exponent from a shallow residue, demonstrating information-completeness.

1 Introduction

1.1 The Question

The Mersenne Prime Mod-144 Signature Theorem (companion paper, “Two Lattices, One Number”) establishes that $2^p - 1$, for prime $p > 3$, is confined to two residues modulo 144, both of which are themselves smaller Mersenne primes. The mechanism is a CRT split: the modulus $144 = 16 \times 9$ separates into a trivial mod 16 component (which stabilizes to 0 once $p \geq 4$) and a mod 9 component governed by the order-6 cyclic group generated by 2.

A natural question follows immediately from the zone-character identification of Part XXIV of the master document (Hard Wall = $\chi_3 = +1$, Temporal = $\chi_3 = -1$ for primes): if $2^p - 1$ is forced into the Hard Wall zone, is there a comparably natural family forced into the *Temporal* zone — the mirror image? This paper answers yes, and identifies that family as classical: the **Fermat numbers**, $F_k = 2^{2^k} + 1$.

1.2 A Second Question

A separate line of inquiry asks whether the lattice’s confinement of M_p to a small residue set is best understood as *lossy* (information about p is discarded, leaving only a coarse classification) or whether, at sufficient depth, the encoding is *lossless*. We show the latter: the residue of M_p modulo a high enough power of 3 determines p completely, and the recovery is computationally easy rather than hard, because the relevant discrete logarithm lives in a group of smooth (Tier-1) order.

2 The Fermat Mod-144 Mirror Theorem

2.1 Why Mersenne and Fermat Should Be Mirrors

Proposition 2.1 (Mod-3 mirror, elementary). *For prime $p > 2$: $M_p = 2^p - 1 \equiv 1 \pmod{3}$. For $k \geq 1$: $F_k = 2^{2^k} + 1 \equiv 2 \pmod{3}$.*

Proof. $2 \equiv -1 \pmod{3}$, so $2^n \equiv (-1)^n \pmod{3}$. For odd p , $2^p \equiv -1 \pmod{3}$, so $M_p = 2^p - 1 \equiv -2 \equiv 1 \pmod{3}$. For $k \geq 1$, 2^k is even, so $2^{2^k} \equiv 1 \pmod{3}$, giving $F_k \equiv 2 \pmod{3}$. $\square$

By the Zone-Character Identity (Theorem 24.1), residues $\equiv 1 \pmod{3}$ are Hard Wall ($\chi_3 = +1$) and residues $\equiv 2 \pmod{3}$ are Temporal ($\chi_3 = -1$). Proposition 2.1 already shows the two families are forced to opposite zones at the mod-3 level; the theorem below sharpens this to the exact residues modulo 144.

2.2 The Theorem

Theorem 2.2 (Fermat Mod-144 Mirror Theorem). *For every $k \geq 2$,*

$$F_k \bmod 144 \in \{17, 113\},$$

with $F_k \equiv 17 \pmod{144}$ when k is even and $F_k \equiv 113 \pmod{144}$ when k is odd. Both residues lie in the Temporal zone.

Proof. Write $144 = 16 \times 9$.

Mod 16. For $k \geq 2$, $2^k \geq 4$, so $2^{2^k} \equiv 0 \pmod{16}$, giving $F_k \equiv 1 \pmod{16}$.

Mod 9. The multiplicative order of 2 modulo 9 is 6. We need $2^k \bmod 6$ for $k \geq 2$: since 2^k is even and $2^k \bmod 6 \in \{2, 4\}$ for $k \geq 1$ (direct check: $2^2 = 4$, $2^3 = 8 \equiv 2$, $2^4 = 16 \equiv 4$, $2^5 = 32 \equiv 2$, alternating), $2^k \equiv 4 \pmod{6}$ for even k and $2^k \equiv 2 \pmod{6}$ for odd k (for $k \geq 2$). Hence:

$$k \text{ even} : 2^{2^k} \equiv 2^4 \equiv 7 \pmod{9}, \quad k \text{ odd} : 2^{2^k} \equiv 2^2 \equiv 4 \pmod{9}.$$

So $F_k \equiv 8 \pmod{9}$ (k even) or $F_k \equiv 5 \pmod{9}$ (k odd) — both in $\{2, 5, 8\}$, the Temporal zone.

Recombination. Solving the CRT system $x \equiv 1 \pmod{16}$, $x \equiv 8 \pmod{9}$ gives $x = 17$; solving $x \equiv 1 \pmod{16}$, $x \equiv 5 \pmod{9}$ gives $x = 113$. $\square$

Verified against all five known Fermat primes ($F_0 = 3$, $F_1 = 5$, $F_2 = 17$, $F_3 = 257$, $F_4 = 65537$) and $F_k \bmod 144$ computed directly for $k = 2, \dots, 24$: zero deviations from $\{17, 113\}$, alternating exactly by parity as proved.

2.3 Small-Case Exceptions

$F_0 = 3$ and $F_1 = 5$ fall outside Theorem 2.2's hypothesis ($k \geq 2$), for the same structural reason the Fibonacci and Mersenne signature theorems exclude their own small cases: the mod-16 component has not yet stabilized to 0 ($2^{2^0} = 2$, $2^{2^1} = 4$, neither divisible by 16). This mirrors §2.3 of “Two Lattices, One Number” precisely.

2.4 Self-Reference

Observation. $17 = F_2$ is itself a Fermat prime. The Fermat signature is self-referential in exactly the sense already established for Fibonacci ($\{1, 5, 13, 89\} = \{F_1, F_5, F_7, F_{11}\}$) and Mersenne ($\{31, 127\} = \{M_5, M_7\}$): every large member of the family, reduced mod Λ , produces a smaller member of the same family. 113 is not itself a Fermat number ($113 \neq 2^{2^k} + 1$ for any integer k); the self-reference here is single rather than double, matching the Mersenne case’s structure ($31 = M_5$ prime, but the pairing is what closes, not every element individually).

2.5 The Complete Mirror Table

Family	Form	Signature mod 144	Zone	Self-reference
Mersenne	$2^p - 1$, p prime > 3	$\{31, 127\}$	Hard Wall	$31 = M_5$, $127 = M_7$
Fermat	$2^{2^k} + 1$, $k \geq 2$	$\{17, 113\}$	Temporal	$17 = F_2$

Table 1: The two classical families occupy disjoint, opposite zones at $\Lambda = 144$, with the sign in $2^n \mp 1$ determining the zone via χ_3 .

3 The 3-adic Reverse Decoder

3.1 Motivation

Theorem 2.2 and the Mersenne signature theorem both express a *restriction*: the exponent’s family is confined to a small residue set modulo 144. This section asks whether that restriction is the shallow edge of something stronger — whether the full residue system, taken to sufficient depth, retains complete information about the exponent, rather than discarding it.

3.2 The Mechanism

Proposition 3.1 (Classical). *2 is a primitive root modulo 3^k for every $k \geq 1$; the group $(\mathbb{Z}/3^k\mathbb{Z})^*$ is cyclic of order $2 \cdot 3^{k-1}$.*

This is standard (Gauss); it is recorded here because of its consequence for this framework specifically: **the group order $2 \cdot 3^{k-1}$ is itself Tier-1** — built only from the primes 2 and 3. A cyclic group of smooth order admits an efficient discrete logarithm (e.g. Pohlig–Hellman), in sharp contrast to the general discrete logarithm problem’s presumed hardness in groups of large prime order.

Theorem 3.2 (Reverse Decoder). *For $p < 2 \cdot 3^{k-1}$, the residue $M_p \bmod 3^k$ uniquely determines p , and p is recoverable in time polynomial in k .*

Proof. By Proposition 3.1, $x \mapsto 2^x \bmod 3^k$ is a bijection $\mathbb{Z}/(2 \cdot 3^{k-1})\mathbb{Z} \rightarrow (\mathbb{Z}/3^k\mathbb{Z})^*$. Given $r = M_p \bmod 3^k$, compute $c = r + 1 \bmod 3^k$; then p is the unique solution of $2^p \equiv c \pmod{3^k}$ with $p < 2 \cdot 3^{k-1}$, found via Pohlig–Hellman in the smooth-order group in $O(\text{poly}(k))$ time. $\square$

3.3 Demonstration

Small case. For $p = 31$, $k = 4$ ($3^4 = 81$, group order $54 > 31$): $M_{31} \bmod 81 = 64$. Solving $2^x \equiv 65 \pmod{81}$ gives $x = 31$. Correct.

World record. The current largest known Mersenne prime has exponent $p = 136,279,841$ (October 2024). Taking $k = 18$ ($3^{18} = 387,420,489$, group order $258,280,326 > p$):

$$M_p \bmod 3^{18} = 156,704,278.$$

Given only this 9-digit number, solving the discrete logarithm recovers $p = 136,279,841$ exactly, in under one millisecond — the full identity of a Mersenne prime with more than 41 million digits, recovered from a residue smaller than a telephone number.

3.4 What This Does and Does Not Establish

Established. The mod-144 (or mod- 3^k) signature is not a lossy classification that discards information about p ; taken to sufficient 3-adic depth, it is a *lossless, efficiently invertible encoding* of p . In this precise sense, the pair $\{2, 3\}$ functions as a matched encoder/decoder: exponentiation by 2 writes the exponent into the residue; the smoothness of 3^k 's unit group, inherited from 3 alone, reads it back out.

Not established. This provides no test for whether a given p yields a *prime* M_p — that remains exactly as hard as before (Lucas–Lehmer or equivalent), and nothing here changes the search for new Mersenne primes. The decoder recovers p from $M_p \bmod 3^k$ when p is already known to be the relevant exponent; it is a statement about information content, not a primality oracle. We state this limitation plainly to foreclose overreading: §4 of “Two Lattices, One Number” already established that the residue signature cannot select promising exponents in advance, and nothing in this paper reopens that question.

4 What This Paper Establishes

5 Open Problems

- OP-1.** Does an analogous mirror theorem hold for the golden lattice — is there a Fibonacci-side family occupying the Temporal zone, opposite the Fibonacci Prime Mod-144 Signature's own zone placement?
- OP-2.** Extend the Reverse Decoder to the Fibonacci and Pell signature theorems: since their governing periods (24, 48) are also Tier-1, do analogous smooth-order discrete-log decoders exist for reconstructing Fibonacci or Pell indices from residues?
- OP-3.** Determine whether the Reverse Decoder's information-completeness has any bearing on the free-verification application of §H.6/“Two Lattices, One Number” — does a deeper (but still cheap) residue check provide a stronger integrity guarantee for large-scale prime search than the shallow mod-144 check alone?
- OP-4.** Search for further classical prime-generating families (Cullen numbers, Woodall numbers, Proth numbers) and test each for a $\Lambda = 144$ signature and zone placement, following the method of §2.

Claim	Status
$M_p \equiv 1 \pmod{3}$, $F_k \equiv 2 \pmod{3}$ (Proposition 2.1)	Proved
Fermat Mod-144 Mirror Theorem: $F_k \bmod 144 \in \{17, 113\}$, $k \geq 2$ (Theorem 2.2)	Proved , verified $k = 2, \dots, 24$ and all 5 known Fermat pri
Mersenne and Fermat occupy opposite, disjoint zones at $\Lambda = 144$	Proved (Proposition 2.1 + Zone-Character Identity)
Fermat signature self-reference ($17 = F_2$)	Verified
2 is a primitive root mod 3^k ; group order is Tier-1 (Proposition 3.1)	Classical , correctly cited
Reverse Decoder: p recoverable from $M_p \bmod 3^k$ (Theorem 3.2)	Proved
Demonstrated on $p = 31$ and the current world record $p = 136,279,841$	Verified , both exact
The signature is information-complete at sufficient depth	Established , precisely scoped
This provides a forward test for Mersenne primality	Not claimed — explicitly false

A Appendix A. The Optimal 3-adic Decoder Depth

Section 3 established that the residue

$$M_p \bmod 3^k$$

contains sufficient information to recover the exponent p , provided the modulus is taken to a sufficiently high power of 3. The natural remaining question is whether this depth is optimal, or whether the same recovery could be achieved with a smaller modulus.

This appendix answers that question completely.

A.1 Theorem A.1 (Minimal Decoder Depth)

Let

$$M_p = 2^p - 1,$$

and suppose the exponent satisfies $p > 0$.

Define

$$k_{\min} = 1 + \left\lceil \log_3 \left(\frac{p+1}{2} \right) \right\rceil.$$

Then:

1. For every $k \geq k_{\min}$, the residue

$$M_p \bmod 3^k$$

determines p uniquely.

2. For every $k < k_{\min}$, uniqueness is impossible: there exists another exponent

$$q \neq p$$

having the same residue modulo 3^k .

Consequently, $k_{\min}$ is the unique minimal decoding depth.

A.2 Proof

The proof follows directly from the multiplicative order of 2 modulo powers of 3.

By Proposition 3.1,

$$\text{ord}_{3^k}(2) = 2 \cdot 3^{k-1}.$$

Therefore

$$2^a \equiv 2^b \pmod{3^k}$$

if and only if

$$a \equiv b \pmod{2 \cdot 3^{k-1}}.$$

Subtracting one from both sides preserves the congruence, giving

$$M_a \equiv M_b \pmod{3^k}$$

if and only if

$$a \equiv b \pmod{2 \cdot 3^{k-1}}.$$

Hence the residue modulo 3^k determines the exponent only modulo

$$2 \cdot 3^{k-1}.$$

If

$$p < 2 \cdot 3^{k-1},$$

then p is the unique representative of its congruence class among all nonnegative integers less than the group order. Therefore the residue uniquely determines the exponent.

Conversely, suppose

$$p \geq 2 \cdot 3^{k-1}.$$

Define

$$q = p - 2 \cdot 3^{k-1}.$$

Since

$$2^{2 \cdot 3^{k-1}} \equiv 1 \pmod{3^k},$$

we have

$$2^p = 2^q 2^{2 \cdot 3^{k-1}} \equiv 2^q \pmod{3^k},$$

and therefore

$$M_p \equiv M_q \pmod{3^k}.$$

Thus two distinct exponents produce the same residue, making unique decoding impossible.

The smallest value of k for which

$$2 \cdot 3^{k-1} > p$$

holds is precisely

$$k_{\min} = 1 + \left\lceil \log_3 \left(\frac{p+1}{2} \right) \right\rceil,$$

which completes the proof. □

A.3 Corollary A.2 (Optimality)

The Reverse Decoder of Theorem 3.2 uses the least possible amount of 3-adic information.

No modulus

$$3^j, \quad j < k_{\min},$$

can uniquely recover the exponent, while every modulus

$$3^j, \quad j \geq k_{\min},$$

can do so.

Thus the decoder is not merely correct—it is provably optimal.

A.4 Example: The Current World Record

The current largest known Mersenne prime has exponent

$$p = 136,279,841.$$

Since

$$2 \cdot 3^{16} = 86,093,442 < 136,279,841 < 258,280,326 = 2 \cdot 3^{17},$$

the minimal decoder depth is

$$k_{\min} = 18.$$

This is exactly the modulus used in the demonstration of Section 3.3.

Therefore the reconstruction of the world-record exponent from

$$M_p \bmod 3^{18}$$

is not merely sufficient—it is mathematically optimal. No smaller power of three can uniquely determine the exponent.

A.5 Interpretation

The Optimal Decoder Theorem sharpens the information-completeness result of Section 3. Rather than asserting only that sufficiently deep 3-adic residues retain complete information about the exponent, it identifies the exact threshold at which information loss disappears.

The quantity

$$2 \cdot 3^{k-1},$$

the order of the unit group modulo 3^k , acts as the lattice’s information capacity. Below this threshold, collisions between exponents are unavoidable; above it, the residue determines the exponent uniquely.

In this precise sense, the Reverse Decoder operates at the information-theoretic minimum: every retained 3-adic digit is necessary, and no redundant depth is required. ““

References

- [1] G. Gurwell, *Two Lattices, One Number: The Fibonacci and Mersenne Prime Signatures of $\Lambda = 144$, and Why This Depth Is Exact*, CTF Framework Research Series, Zenodo, 2026.
- [2] G. Gurwell, *Prime Lattice Coherence Framework: Master Document, Part XXIV — The Zone Character*, CTF Framework Research Series, Zenodo, 2026.
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- [4] S. Pohlig, M. Hellman, “An Improved Algorithm for Computing Logarithms over $\text{GF}(p)$ and Its Cryptographic Significance,” *IEEE Transactions on Information Theory*, 24(1), 106–110, 1978.
- [5] Great Internet Mersenne Prime Search (GIMPS), current record $M_{136,279,841}$, October 2024.

All arithmetic verified by exact integer computation (Python, sympy for primality and discrete logarithm); the world-record demonstration uses the publicly documented exponent, not a newly computed value. Verification code at ctftheory.com.

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